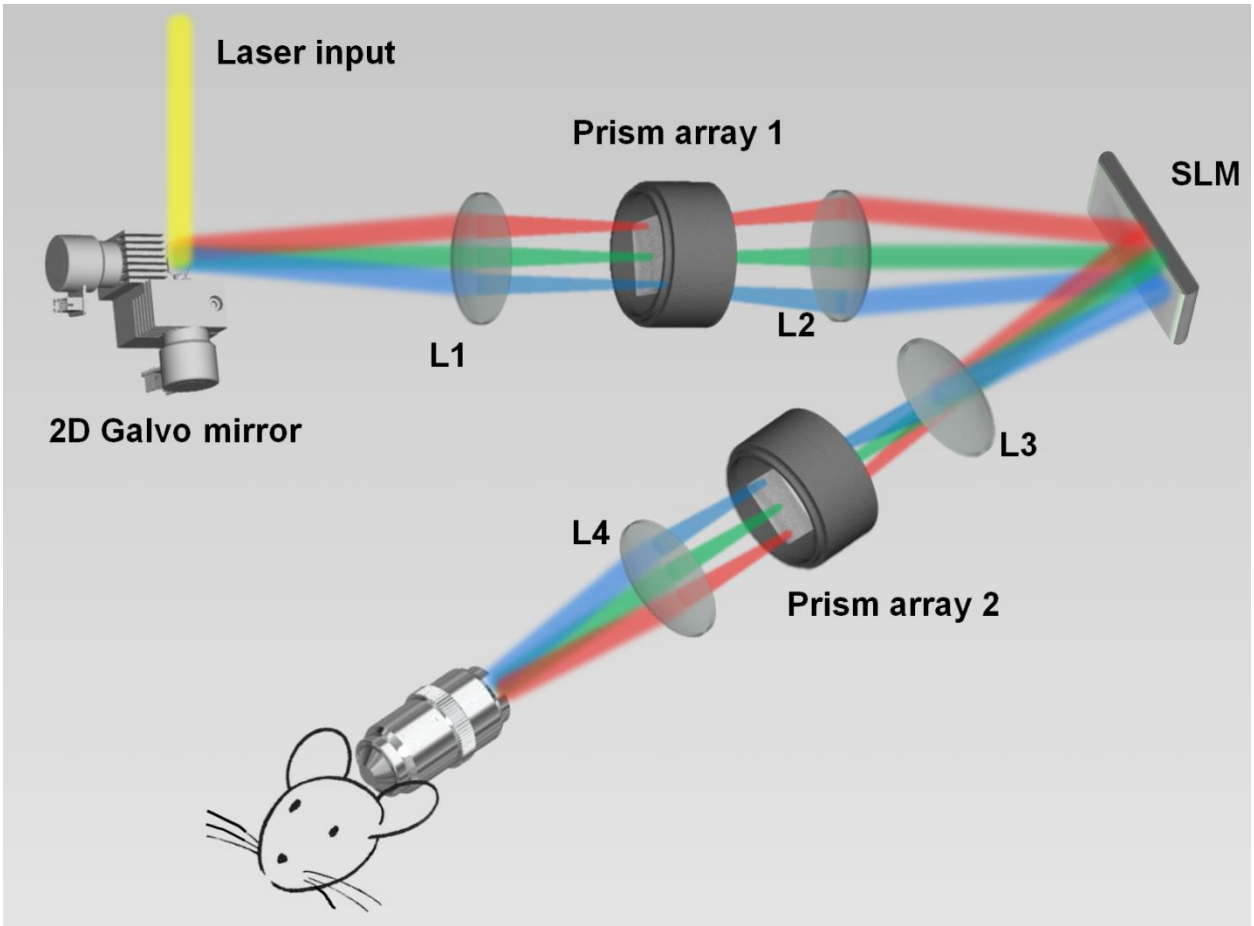
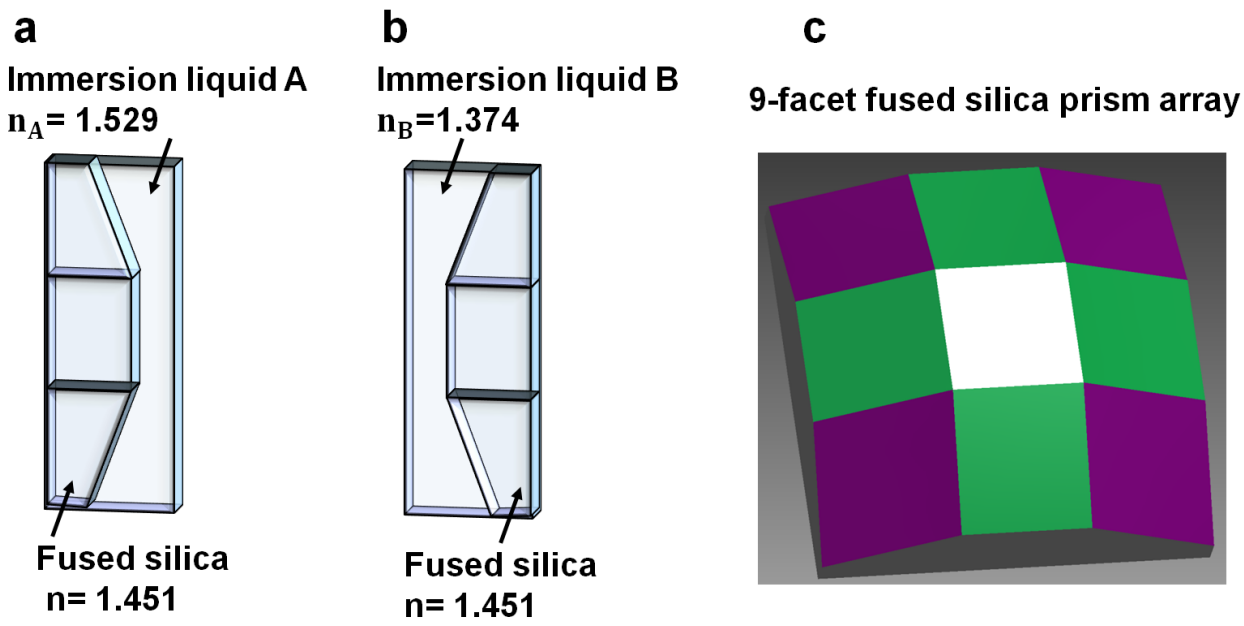



**Supplementary Figure 1**

**Design of MPAO based laser scanning two-photon fluorescence microscope.**

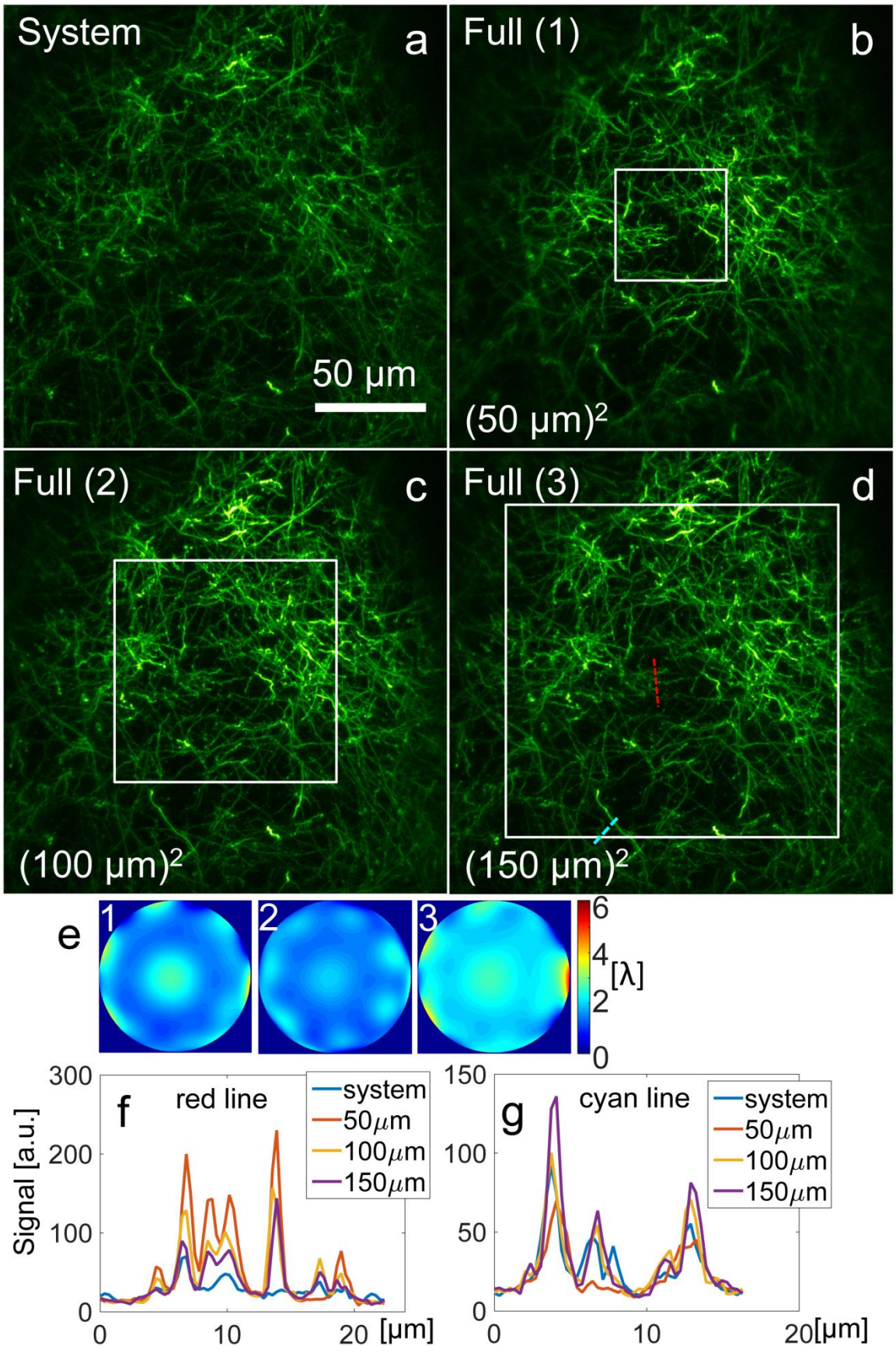
Different from conventional AO based microscopes, we inserted two prism arrays at the focal planes between the two pairs of relay lenses, which generated a two-dimensional array of pupil images on the SLM. The Galvo mirror aperture was 5 mm in diameter, which was demagnified in the ratio of 11:15 by L1 and L2 and then magnified 50:11 by L3 and L4. The beam diameter at the pupil plane of the objective lens (Nikon 25× NA 1.1, full pupil 17.6 mm) was 16.7 mm, resulting in an effective excitation NA of 1.04.




**Supplementary Figure 2**

**Design of the tunable prism array.**

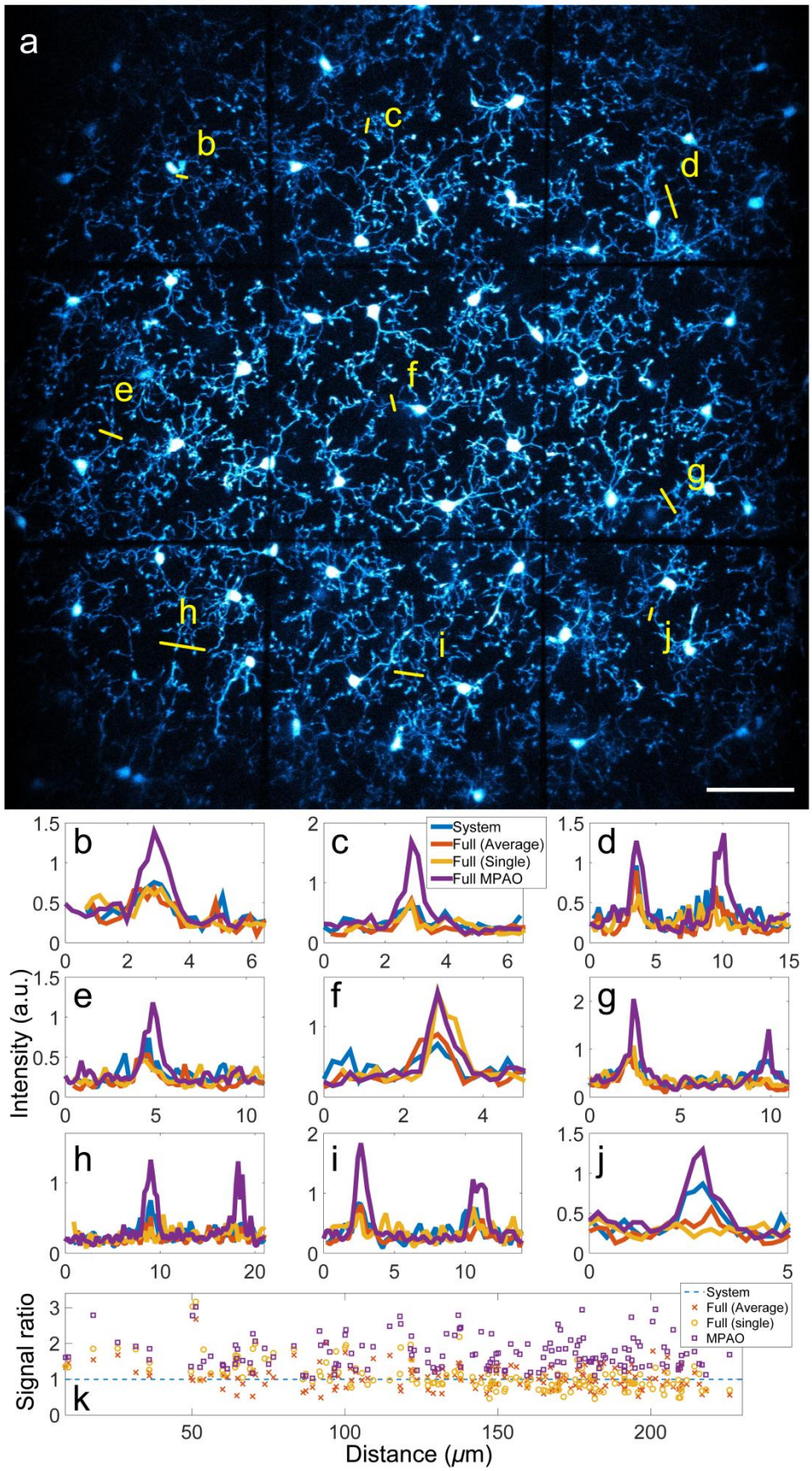
(**a**, **b**) We immersed the fused silica prism array in refractive index matching liquid (Cargille, Laser liquid, code 1074 and 3421). Controlling the refractive index of the immersion media allowed us to adjust the beam tilting angles. By using two liquids of different refractive indices (one higher and one lower than that of fused silica), we formed the complementary prism arrays. The angle tilt caused by one prism array was canceled by the other array. (**c**) 3D rendering of the 3×3 prism array. The size of each facet of the prism is 9.375 × 9.375 $mm^2$. The angle between the purple facets (corners) and the central facet is 31.5 degree and the angle between the green facets and the central facet is 23.4 degree. The central thickness of the prism array is 10 mm.


System
a
50 μm
Full (1)
b
(50 μm)²
Full (2)
c
(100 μm)²
Full (3)
d
(150 μm)²
e
1
2
3
6
4
2
0
[λ]
f
red line
system
50μm
100μm
150μm
Signal [a.u.]
300
200
100
0
0
10
20
[μm]
g
cyan line
system
50μm
100μm
150μm
150
100
50
0
0
10
20
[μm]

**Supplementary Figure 3**

**Wavefront measurement using image areas of different sizes as feedback.**

As biological tissue is inhomogeneous, the aberration is position-dependent. Therefore, optimizing the wavefront within a small region could yield high-quality correction albeit with a small effective correction FOV. In comparison, optimizing the wavefront over a large region could increase the correction FOV albeit with reduced correction quality. Here we compare the *in vivo* imaging results obtained by using system correction (**a**) and by using 50 × 50 (**b**), 100 × 100 (**c**), 150 × 150 (**d**) $\mu m^2$ area as the feedback during Zernike mode based wavefront optimization at 50 μm depth under the dura of Thy1-YFP mice. For these measurements, we used conventional AO microscopy (not MPAO) and the Full correction compensates both system aberration and sample aberration. The imaging FOV was 200 × 200 $\mu m^2$ and the sample aberration wavefront of each measurement is also shown (**e**). The solid white boxes indicate the regions which were used as the feedback during wavefront optimization. (**f**) Intensity plot across the red line in **d**. (**g**) Intensity plot across the cyan line in **d**. Experimentally, we chose 150 × 150 $\mu m^2$ area as the feedback area and as the image segment size in MPAO as a compromise between the correction quality and the effective correction FOV. By using prism arrays of different sizes, we can switch between different segment sizes for different biological tissues in MPAO (see Supplementary Note 4). The data shown in this figure is representative of two experiments.

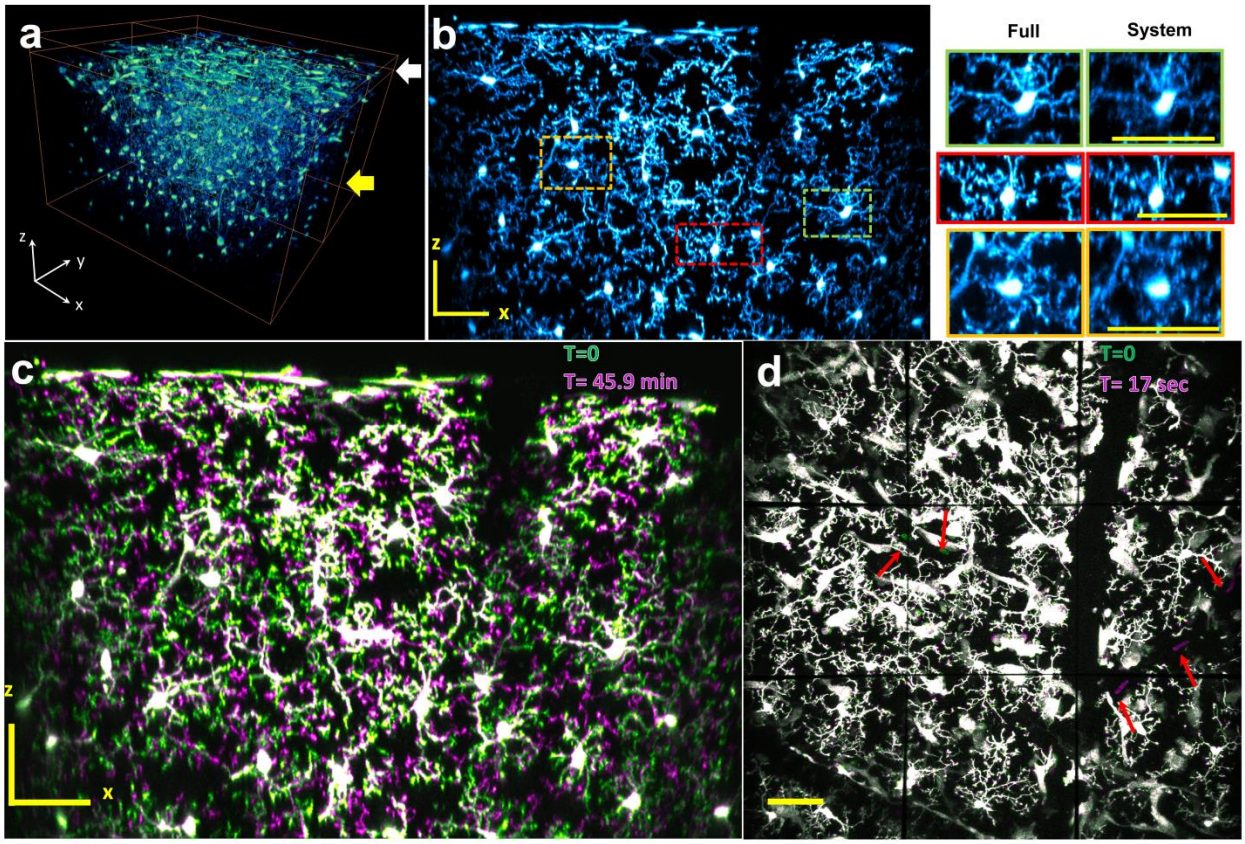



Nature Methods: doi:10.1038/nmeth.4290

**Supplementary Figure 4**

**Quantitative comparison between MPAO and conventional AO methods.**

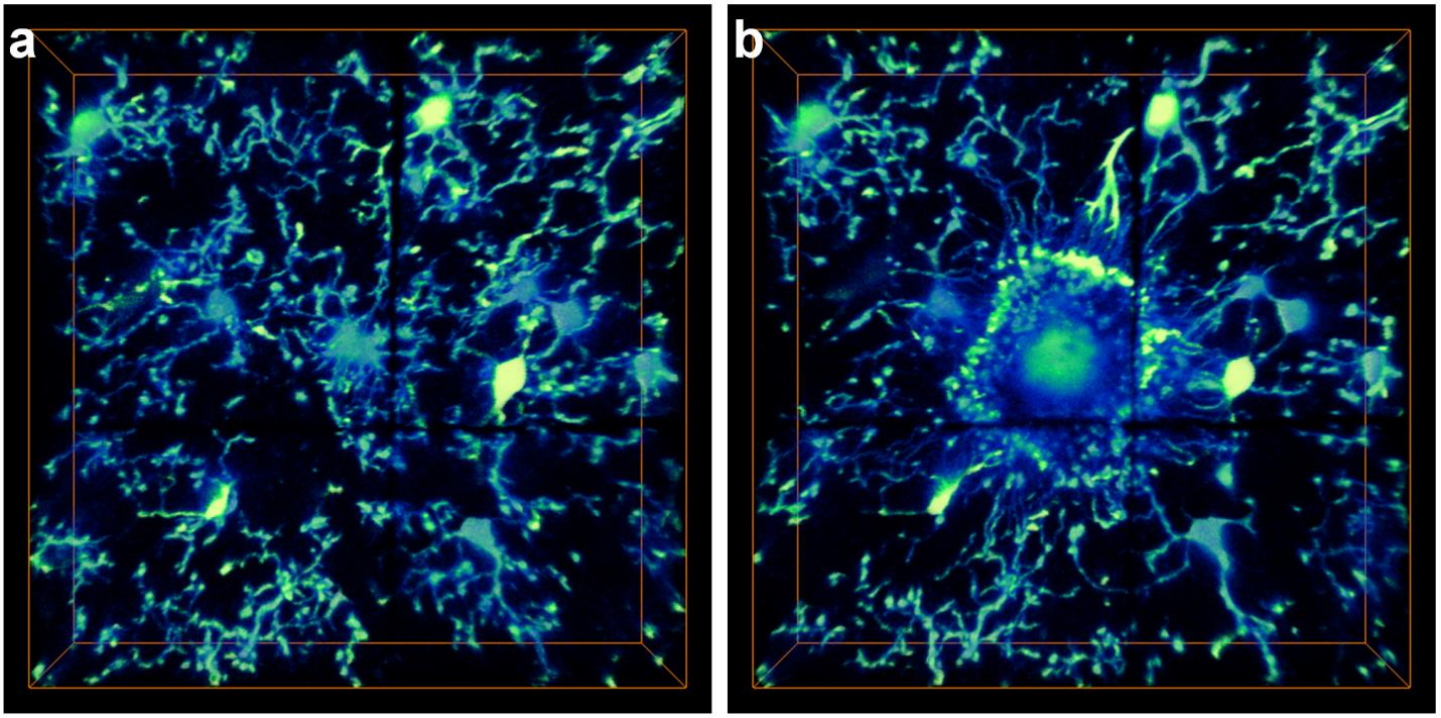
(**a**) Maximum intensity projection of the 3D volume shown in Fig. 1. The yellow lines indicate the positions of cross-section plot (**b**-**j**). (**k**) We compare MPAO with Full average (conventional AO using the entire image as feedback during wavefront measurement) and Full single (conventional AO using the central image segment as feedback during wavefront measurement). The signals are normalized by the system correction signal values. The data shown in this figure is representative of three experiments.




**Supplementary Figure 5**

**High-resolution large-volume imaging of resting state microglia.**

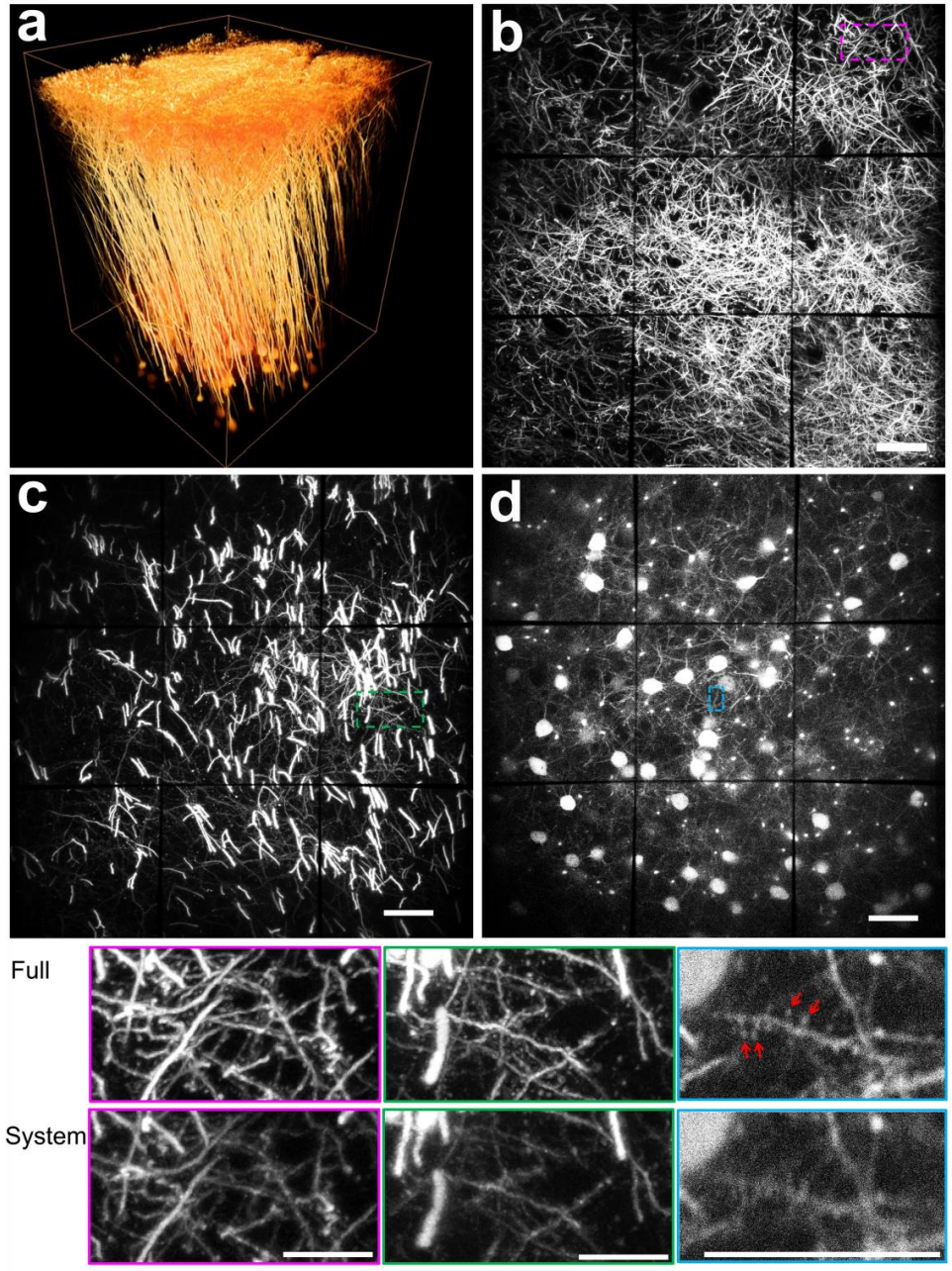
(**a**) 3D rendering of microglia recorded by MPAO within a 450 × 450 × 300 $\mu m^3$ volume. (**b**) MIP of a 20 μm thick *x-z* image stack (the vertical plane marked by the yellow arrow in **a**). Zoomed-in views show the comparison of full correction and system correction. (**c**) Long-term dynamic imaging of the vertical stack shown in **b**. We overlay two stacks recorded at different time (*t* = 0 and 45.9 min). (**d**) Imaging fast dynamics within a 50 μm thick *x-y* image stack (the horizontal plane marked by the white arrow in **a**). The GFP-expressing leukocytes (marked by red arrows) trafficking through brain vasculatures were recorded during the time-lapse recording (Supplementary Video 2). Scale bar: 50 μm. The data shown in this figure is representative of two experiments.




**Supplementary Figure 6**

**3D rendering of the tissue volume around the ablation site.**

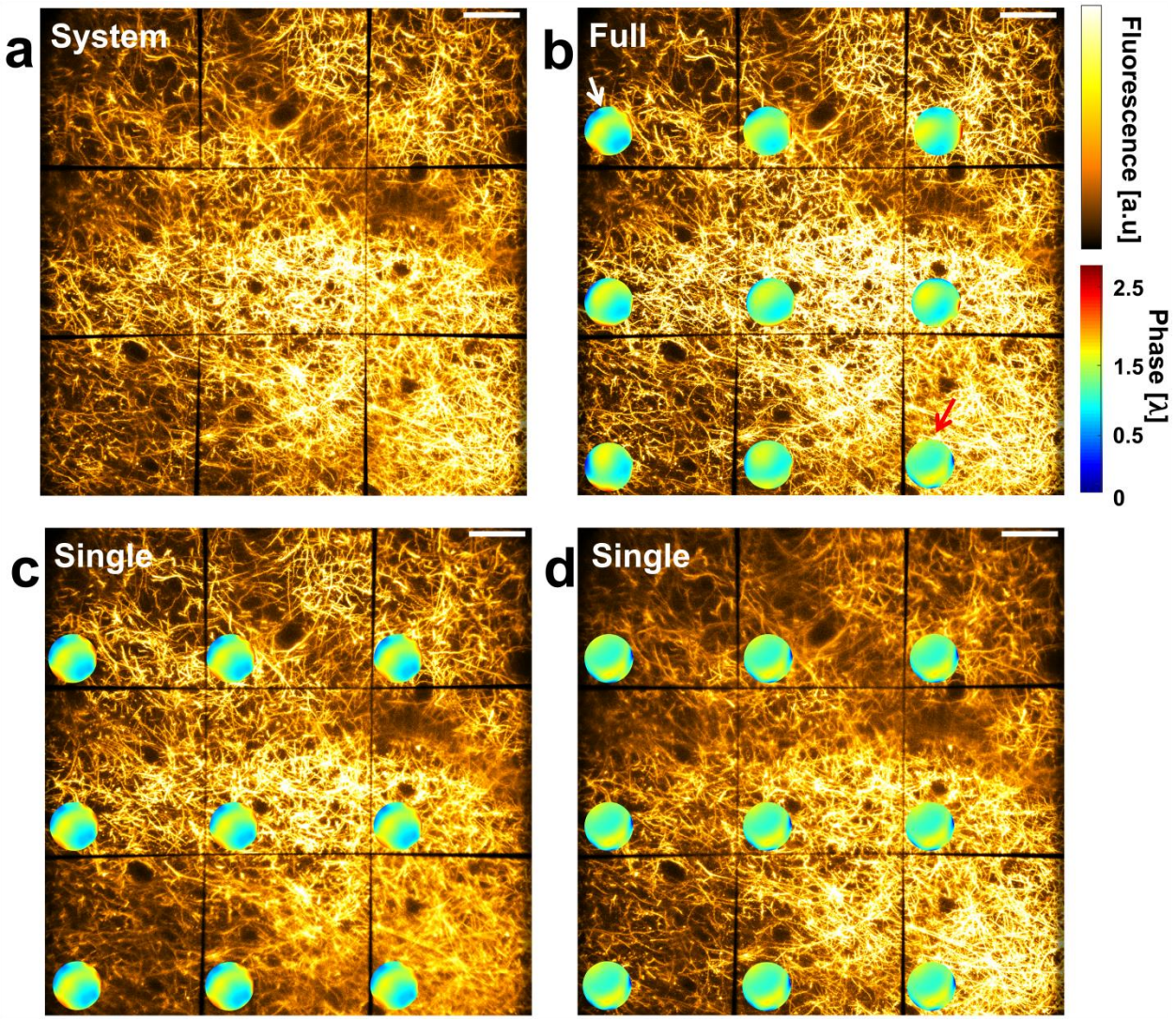
(**a**) Resting state microglia before laser ablation. (**b**) Activated microglia after laser ablation. Volume size: 166 × 166 × 35 $\mu m^3$. The dynamics is shown in Supplementary Video 3. The data shown in this figure is representative of three experiments.




**Supplementary Figure 7**

**High-resolution large-volume deep-tissue imaging of neurons.**

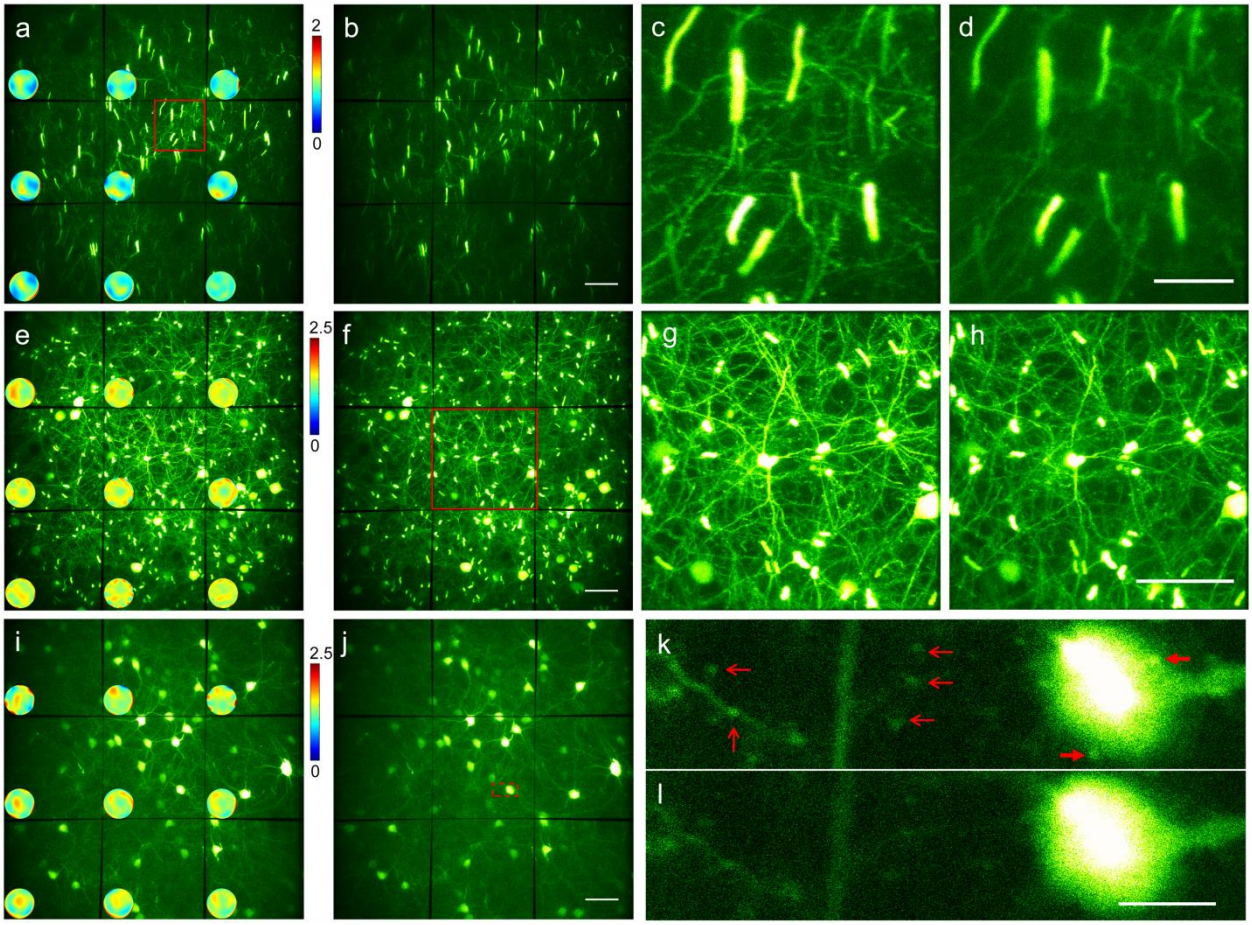
(**a**) 3D rendering of neurons recorded by MPAO within a 450 × 450 × 550 μm$^3$ volume. (**b**, **c**) MIPs along *z* of 50 μm thick image stack centered at the depth of 75 and 325 μm, respectively. (**d**) The *x*-*y* image at 623 μm under the dura. The insets show the zoomed-in view to compare the images with full (tissue aberration + system aberration) and with system wavefront correction at different positions and depths. Red arrows, spines on basal dendrites of layer 5 neurons. Scale bars: 50 μm in **b**-**d**, 20 μm in the zoomed-in view. The data shown in this figure is representative of three experiments.




**Supplementary Figure 8**

**Position-dependent aberrations at the shallow layer of mouse neocortex.**

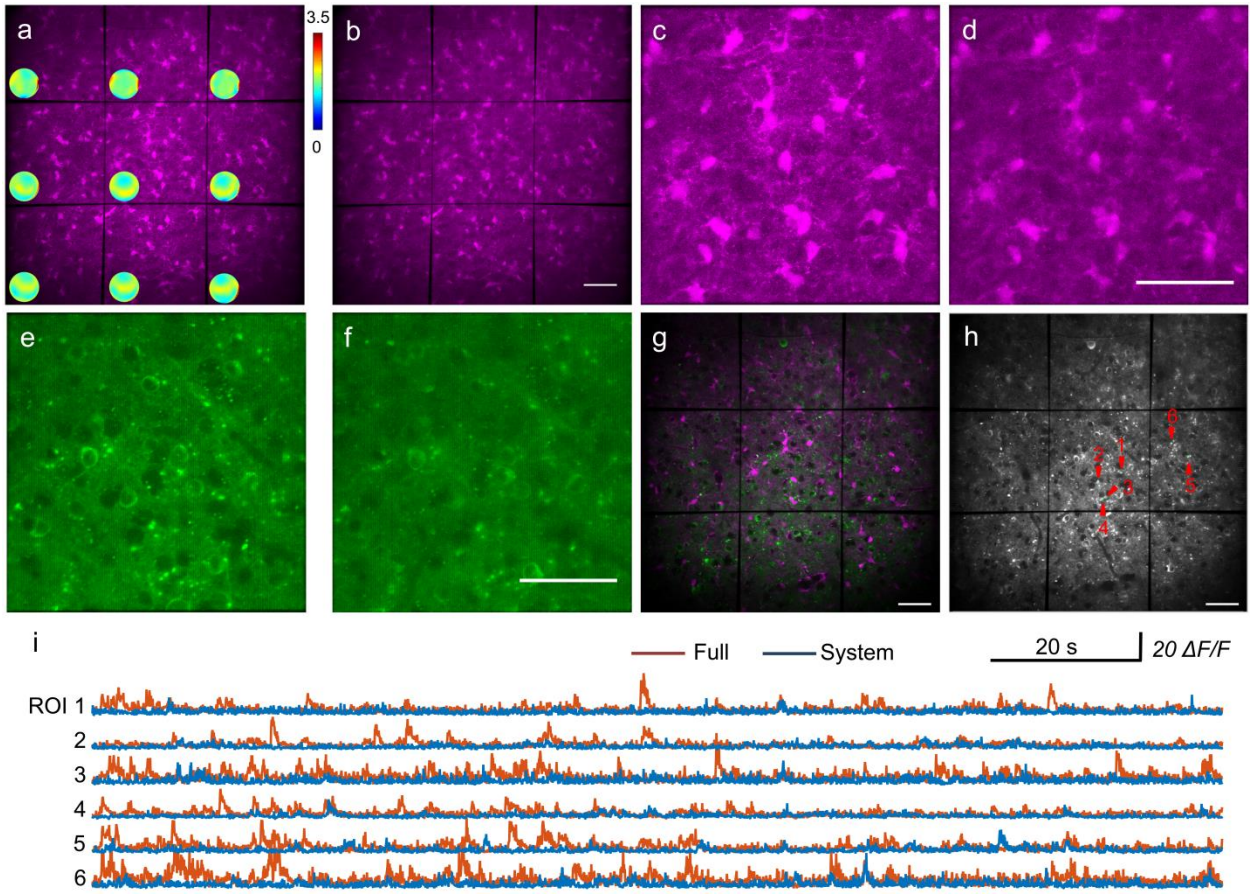
MIPs along *z* axis of 10 μm thick tissue volume at a depth of 75 μm acquired with system correction (**a**), full correction (**b**), single correction using the wavefront marked by the white arrow (**c**) and the red arrow (**d**) in **b**. The wavefront profiles are shown, with the amplitude scaled to the laser wavelength of 935 nm. Scale bar: 50 μm. The data shown in this figure is representative of two experiments.




**Supplementary Figure 9**

**Deep imaging of neural dendrites and spines.**

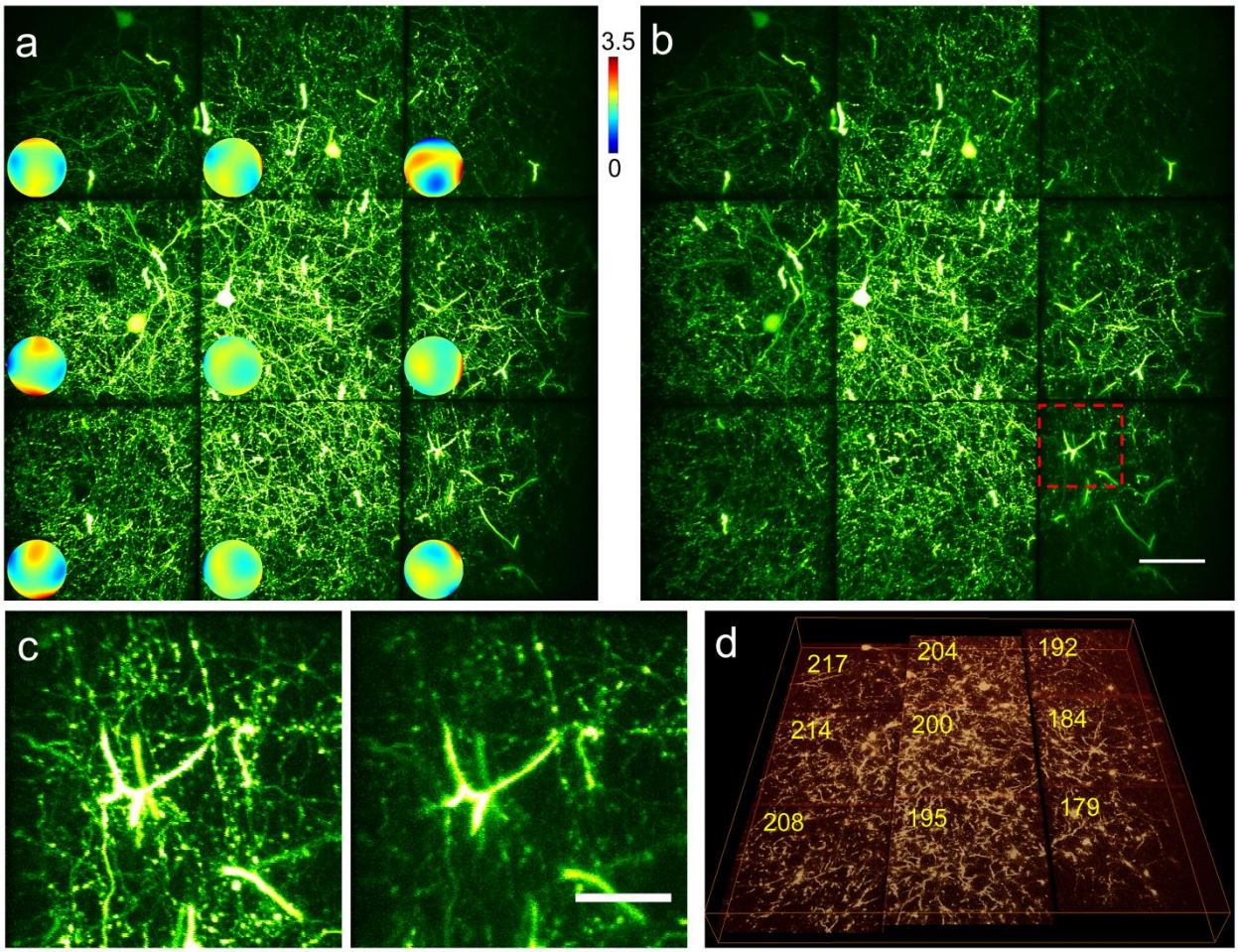
(**a**, **b**) MIPs along *z* axis of the tissue volume at 430-470 μm under the dura with full and with system correction, respectively. The wavefront profiles for the nine regions measured at the depth of 450 μm are shown in **a**. The wavefront amplitude is scaled to the laser wavelength of 935 nm. Scale bar: 50 μm. (**c**, **d**) The zoomed-in view of the area enclosed by the red box in **a** with full and with system correction, respectively. Scale bar: 20 μm. (**e**, **f**) MIPs along *z* axis of the tissue volume at 510-530 μm under the dura with full and with system correction, respectively. The wavefront profiles for the nine regions measured at the depth of 520 μm are shown in **e**. Scale bar: 50 μm. (**g**, **h**) The zoomed-in view of the area enclosed by the red box in **f** with full and with system correction, respectively. Scale bar: 50 μm. (**i**, **j**) MIPs along *z* axis of the tissue volume at 630-670 μm under the dura with full and with system correction, respectively. The wavefront profiles measured at the depth of 650 μm are shown in **i**. Scale bar: 50 μm. (**k**, **l**) The zoomed-in view of the area enclosed by the red box in **j** with full and with system correction, respectively. Thin arrows: the spines on the basal dendrites; thick arrows: neurites around the soma; scale bar: 10 μm. The data shown in this figure is representative of three experiments.




**Supplementary Figure 10**

**Calcium imaging of neurons at a depth of 350 μm with MPAO.**

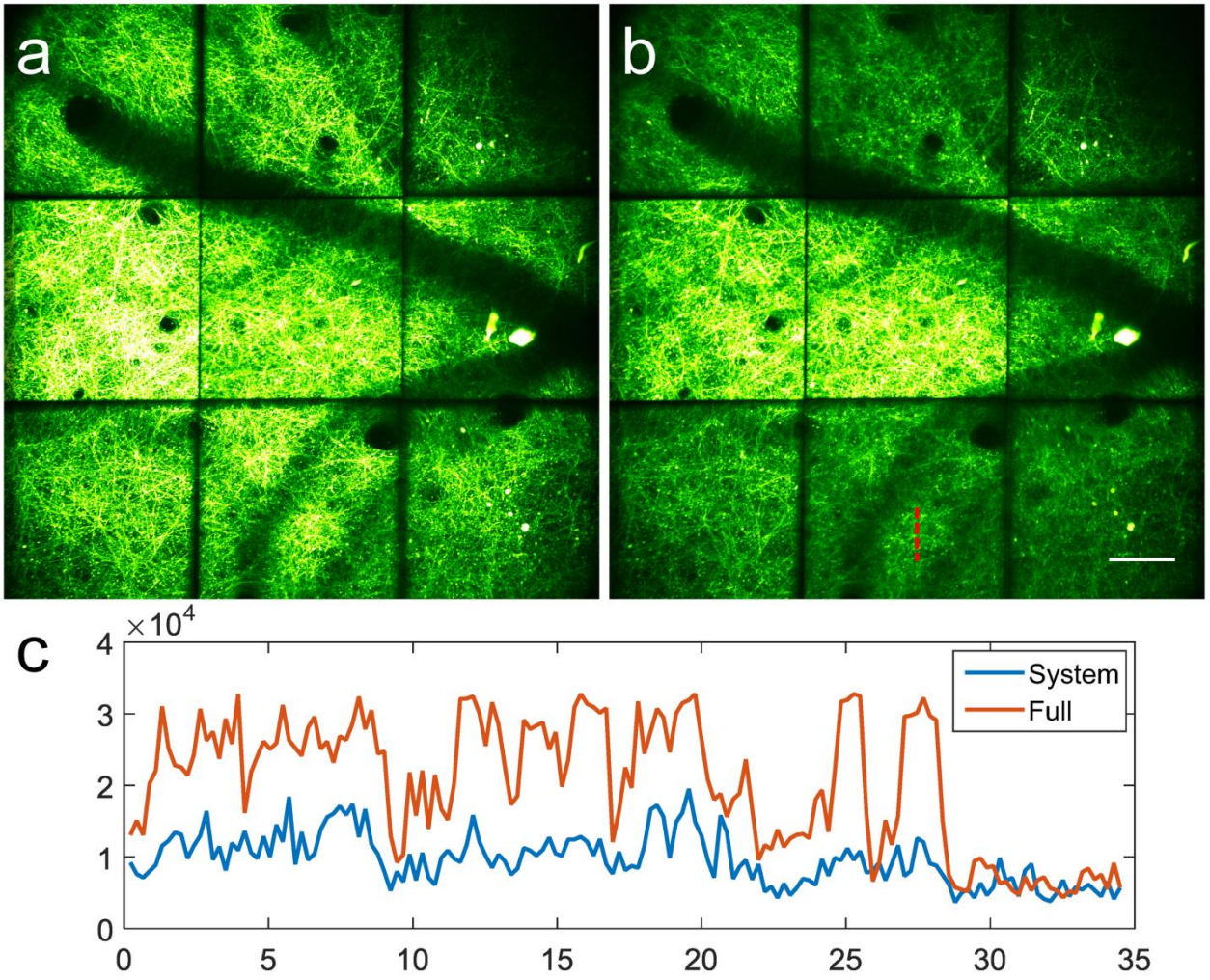
(**a**, **b**) MIPs along *z* of SR101 stained astrocytes at 336-365 μm under the dura with full and with system correction, respectively. The wavefront profiles for the nine regions measured at the depth of 350 μm are shown in **a**. The wavefront amplitude is scaled to the laser wavelength of 935 nm. (**c**, **d**) Zoomed-in view of the central 150 × 150 $\mu m^2$ area in **a** and **b**. (**e**, **f**) Standard deviation of the time-lapse images of GCaMP6s-expressing neurons with full and with system correction, respectively. **c-f** are from the same area. (**g**) Composite view of astrocytes (magenta) and neurons (green) at a depth of 350 μm. (**h**) ROIs for computing calcium transients. (**i**) Calcium transients with full and with system correction. Part of the calcium transient is also shown in Supplementary Video 5. Scale bars: 50 μm. The data shown in this figure is representative of four experiments.




**Supplementary Figure 11**

**MPAO enabled nonplanar imaging.**

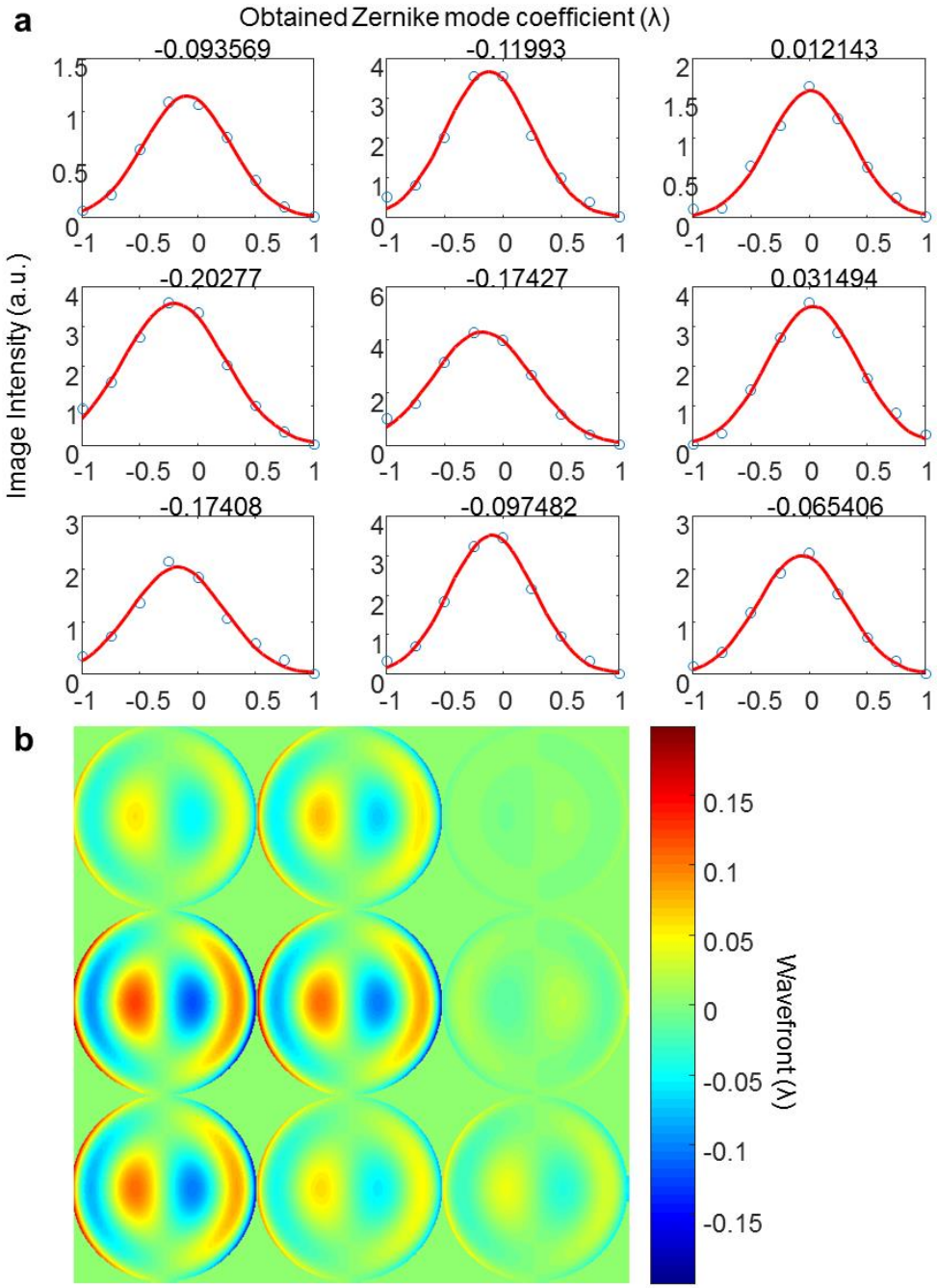
(**a**, **b**) MIPs along *z* axis of the images acquired with full and with system correction, respectively. Each of the image segments was from a 50 μm thick image stack at a different depth. The tissue induced wavefront aberration is shown in **a**. The wavefront amplitude is scaled to the laser wavelength of 935 nm. Scale bar: 50 μm. (**c**) The zoomed-in view of the area enclosed by the red dashed box in **b** with full and with system correction, respectively. Scale bar: 20 μm. (**d**) The 3D distribution of the central planes of the nine image segments (Supplementary Video 6). We show the depth in micron for each plane. The data shown in this figure is representative of three experiments.




**Supplementary Figure 12**

**Signal improvement by MPAO in nonplanar imaging.**

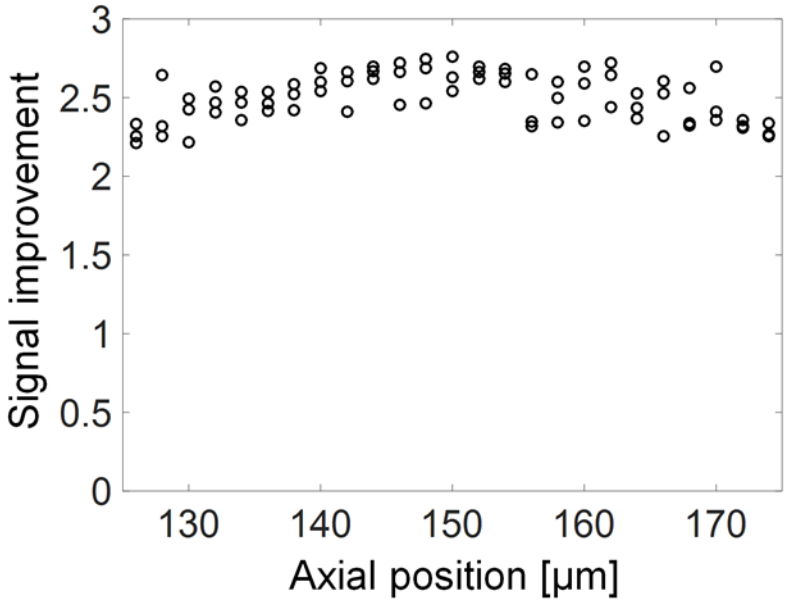
(**a**, **b**) The maximum intensity projection of the volume across 60 μm in depth (6-65 μm under dura), with full correction and system correction, respectively. (**c**) Signal intensity along the red dash line in **b**. The data shown in this figure is representative of three experiments.




**Supplementary Figure 13**

**Zernike mode optimization results for one mode used in Fig. 1.**

(**a**) The image intensity variation for the nine image segments as a function of the applied Zernike mode. The Gaussian fitting returns the center position, which is the needed coefficient for this applied Zernike mode. (**b**) After the measurement, we add the corresponding amount of the measured Zernike mode to the SLM. The entire process repeats for each mode and the final wavefront displayed on the SLM is the summation of all the measured Zernike modes.




**Supplementary Figure 14**

**Dependence of wavefront correction quality on the axial deviation from the measurement plane.**

For volumetric imaging, we measured the wavefront every 50 μm in depth. After the wavefront measurement, we carried out imaging within ± 25 μm from the wavefront measurement plane. Here we show the signal improvement (the ratio of Full correction signal to System correction signal) measured *in vivo* from the dendrites of Thy1-YFP mice at 125-175 μm depth. The wavefront measurement plane was at the center of the image stack (150 μm deep). The data shown in this figure is representative of three experiments.

**Supplementary Table 1: Imaging parameters**

| | Image volume ($\mu m^3$) /field ($\mu m^2$) | Depth ($\mu m$) | Power (mW) | Pixel rate (pixels/s) | Frame rate (Hz) | Number of average | Note |
|---|---|---|---|---|---|---|---|
| Figs. 1d and 1e | 450 × 450 × 40 | 331-370 | 54 | 32 M | 7.65 | 4 | MIP along z |
| Fig. 1i | Zoom-in view of 225 × 225 | 291 | 54 | 32 M | 7.65 | 4 | Data shown: 106 × 106 $\mu m^2$ |
| Fig. 1k | Zoom-in view of 450 × 450 × 35 | 331 - 365 | 90 | 32 M | 7.65 | 3 | MIP along z, data shown: 221 × 221 $\mu m^2$ |
| Figs. 2a and 2b | 450 × 450 × 30 | 436 - 465 | 125 | 32 M | 7.65 | 10 | MIP along z |
| Fig. 2i | 450 × 450 | 450 | 125 | 15.9 M | 15.13 | 1 | / |
| Fig. 3a | 450 × 450 × 60 | 5-65 | 21 | 32M | 7.65 | 6 | Depth color-coded |
| Fig. 3b and 3c | 450 × 450 | 30-53 | 21 | 32M | 7.65 | 1 | Nonplanar Imaging |
| Supp. Figs. 5b and 5c | 450 × 20 × 300 | 1 - 300 | 31 - 103 | 15.9 M | 15.13 | 3 | MIP along y |
| Supp. Fig. 5d | 450 × 450 × 50 | 1 - 50 | 31 - 38 | 15.9 M | 15.13 | 3 | MIP along z |
| Supp. Fig. 6 | Zoom-in view of 450 × 450 × 35 | 331 - 365 | 90 | 32 M | 7.65 | 3 | Data shown: 166 × 166 × 35 $\mu m^3$ |
| Supp. Fig. 7a | 450 × 450 × 550 | 1 - 550 | 29 - 160 | 32 M | 7.65 | 3 | / |
| Supp. Fig. 7b | 450 × 450 × 50 | 51 - 100 | 35 - 43 | 32 M | 7.65 | 3 | MIP along z |
| Supp. Fig. 7c | 450 × 450 × 50 | 301 - 350 | 96 - 116 | 32 M | 7.65 | 3 | MIP along z |
| Supp. Fig. 7d | 450 × 450 | 623 | 125 | 32 M | 7.65 | 6 | / |
| | Inset: Zoom-in view of 90 × 90 | 623 | 160 | 32 M | 7.65 | 6 | Data shown: 34 × 19 $\mu m^2$ |
| Supp. Fig. 8 | 450 × 450 × 10 | 71 - 80 | 38 - 40 | 32 M | 7.65 | 3 | MIP along z |
| Supp. Figs. 9a and 9b | 450 × 450 × 40 | 431 - 470 | 38 | 32 M | 7.65 | 10 | MIP along z |
| Supp. Figs. 9e and 9f | 450 × 450 × 20 | 511 - 530 | 125 | 32 M | 7.65 | 6 | MIP along z |
| Supp. Figs. 9i and 9j | 450 × 450 × 40 | 631 - 670 | 160 | 32 M | 7.65 | 3 | MIP along z |
| Supp. Figs. 9k and 9l | Zoom-in view of 112.5 × 112.5 | 653 | 200 | 32 M | 7.65 | 10 | Data shown: 62 × 15 $\mu m^2$ |
| Supp. Figs. 10a and 10b | 450 × 450 × 30 | 336 - 365 | 38 | 32 M | 7.65 | 6 | MIP along z |
| Supp. Fig. 10i | 450 × 450 | 350 | 125 | 15.9 M | 15.13 | 1 | / |
| Supp. Figs. 11a and 11b | 450 × 450 depth range: 50 | See note | 38 - 54 | 32 M | 7.65 | 6 | MIP along z, central depth is indicated in Supp. Fig. 9d |
| Supp. Fig. 12 | 450 × 450 × 23 | 30-53 | 21 | 32 M | 7.65 | 6 | Nonplanar imaging |
| Supp. Video 1 | 450 × 450 × 40 | 331 - 370 | 54 | 32 M | 7.65 | 4 | MIP along z, same as Figs 1d-1e |
| Supp. Video 2 | 450 × 450 × 50 | 1 - 50 | 31 - 38 | 15.9 M | 15.13 | 3 | MIP along z |
| Supp. Video 3 | 450 × 450 × 35 | 331 - 365 | 90 | 32 M | 7.65 | 3 | MIP along z, same as Fig. 1k |
| Supp. Video 4 | 450 × 450 × 550 | 1 - 550 | 29 - 160 | 32 M | 7.65 | 3 | Same as Supp. Fig. 7a |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Supp. Video 5 | 450 × 450 | 350 | 125 | 15.9 M | 15.13 | 1 | Same as Supp. Fig. 10i |
| Supp. Video 6 | 450 × 450 × 80 | 176 - 220 | 38 - 54 | 32 M | 7.65 | 3 | For data in magenta |
| Supp. Video 7 | 450 × 450 × 23 | 30-53 | 21 | 32 M | 7.65 | 1 | Nonplanar imaging |

**Supplementary Note 1: Comparison of MPAO with other large FOV AO methods**

The purpose of MPAO is to achieve simultaneous wavefront correction over a large imaging FOV. Before the development of MPAO, we have explored several different approaches.

The simplest approach is to measure the averaged wavefront over a large FOV. This can be done by either averaging the wavefront acquired from multiple sub-FOV regions or directly using the signal from a large FOV as the feedback during the wavefront measurement. However, the benefit of AO vanishes for inhomogeneous biological tissues.

An apparent solution is to measure the wavefront of each small region, switch wavefront before imaging the next region, and digitally stitch the image to form a large imaging FOV (e.g. see Ref. 11). However, such an approach has two drawbacks. One is that it is not compatible with many parallel imaging systems such as wide field imaging or multiple-beam scanning microscopy (e.g. spatiotemporal multiplexing). The other is that it slows down the imaging process even for single beam laser scanning microscopy. The widely used resonant Galvo scanner achieves its highest throughput only at large scanning angles. The mechanical scanning favors the long path to achieve the highest scan speed. Thus breaking down a large FOV into multiple small segments and then imaging one by one will inevitably decrease data throughput. Due to this limitation, this approach is not ideal for high-throughput imaging of biological dynamics.

A more sophisticated and elegant approach is the conjugate AO developed in astronomy (e.g. see Ref. 15). However, to make conjugate AO scalable, it requires multiple deformable mirrors and complicated optical and control systems, which may hinder its broad usage for practical bioimaging applications.

Another possibility is to have a wavefront switching system that can operate at high speed to keep up with the laser scanning microscopy (while keeping scanning as fast as possible). Current state-of-the-art wavefront modulators can achieve tens of kilo Hertz update rate. In other words, these methods require tens of microseconds to transit from one wavefront to the other. With the commonly used 8 kHz resonant Galvo, the line scan time is ~60 microseconds. However, if wavefront switching at such a speed is used for practical imaging, there will be no time left for any imaging. For example, if we switch the wavefront three times per line, all the available time will be fully occupied by the wavefront switching process. As a result, the actual imaging duty cycle becomes zero. To have a practical imaging system with high duty

cycle, we would need a wavefront switching system with microsecond or less transition time. To the best of our knowledge, there is still no demonstration of using such wavefront switching for high-throughput *in vivo* imaging.

Through the comparison with other technical possibilities, MPAO has the following major advances.

1, MPAO provides simultaneous wavefront correction which is compatible with both parallel (e.g. wide field, light sheet, light field, PALM/STORM, spatiotemporal multiplexing based multiple-beam scanning) and sequential image acquisitions (e.g. single beam scanning method).

2, MPAO is simple. MPAO differs from conventional AO microscope only in inserting the prism array between the relay lenses.

3, MPAO is scalable. We demonstrated 3×3 arrays in this work. Larger number of arrays (e.g. 6×6) can be implemented to achieve even greater improvement in both FOV and imaging throughput. Even with 600×600 pixels in a typical SLM, we can have 100×100 pixels for each segment in the 6×6 configuration, which are more than sufficient for even high-order wavefront correction.

4, MPAO is compatible with many wavefront measurement methods. Regardless of the measurement method and the complexity of the wavefront, MPAO enables simultaneous wavefront correction over multiple regions.

5, MPAO enables nonplanar microscopy, which can find broad applications in imaging of 3D biological dynamics, such as neurovascular regulation and neural network activity.

## Supplementary Note 2: Overall system design consideration

The overall system design is largely determined by the nature of the biological system of interest. First, we need to evaluate the effective FOV under single wavefront correction (conventional AO). To obtain the optimal size of the FOV, we experimentally varied (gradually increasing) the size of the image which was used for wavefront measurement (Supplementary Fig. 3). For *in vivo* imaging of mouse brain, we found experimentally that 150 micron provided a good compromise between FOV and wavefront correction quality, which was then selected as the image segment size. Next, we determined the overall FOV (how many segments would be needed). We prefer a FOV as large as possible, which is however limited by the capability of the imaging hardware. For example, the laser scanning hardware in our system can operate at a 16 kHz line rate and a 32 M samples/sec throughput (~ two laser pulses for each image pixel). With the spatial resolution and the throughput considered, using more than three elements per line will challenge the available data points per line for satisfying the sampling theorem. Therefore, we chose to implement the 3×3 MPAO system. We selected the focal length ratio of the first relay lens pair (15:11) such that the 9 pupils can fit within the available SLM area. We selected the focal length ratio of the second relay lens pair (11:50) such that the recombined pupil matched well with the water dipping objective lens. Using Zemax, we designed the relay lens pairs based on off-the-shelf achromatic lenses. The design yielded the size and laser beam divergence at the intermediate imaging plane where the prism arrays would be positioned, which further determined the dimension of the prism array and the angular tilt (equal to the beam divergence). For its low optical dispersion and low refractive index, we selected fused silica as the material for prism fabrication. We chose the refractive index of the immersion oil (one higher and one lower than that of fused silica) to provide the complementary (one positive and one negative) angular tilt. The refractive index oil immersion design comes with several advantages. First, it lowers the prism fabrication precision requirement. With the immersion, the surface quality, flatness and angular precision can all accept larger tolerance. Second, the immersion design allows tunability. By changing the refractive index of the immersion oil, we can adjust the beam tilting angle. Third, we can use the same prism array to achieve both positive and negative tilting, which simplifies the fabrication and lowers the fabrication cost. The boundaries between the prism facets caused dark edges, visible in the image. The transition time for the laser beam to move across the boundary is ~0.6 microseconds (min = 0.37; max = 1.11 microseconds), which is ~1% of the 60 microseconds line scan time. This can be further reduced by using relay lenses of greater focal length. On the one hand, the longer focal length means that the laser spot at the prism array plane will have larger size. As a result, the boundary will become relatively smaller compared to the laser beam. On the other hand, the longer focal length means smaller beam divergence. Thus, the needed tilting angle will be reduced, which can reduce the overall prism thickness and further reduce the boundary effect.

## Supplementary Note 3, Quantitative comparison with conventional AO methods

To quantitatively compare MPAO with conventional AO methods, we analyzed the volumetric data from *in vivo* imaging of microglia. Specifically, we compare the signal intensity obtained by MPAO, Full average (using the entire FOV as feedback during wavefront measurement), Full single (using only the central image segment as feedback during wavefront measurement) and system correction. The results are summarized in Supplementary Fig. 4. From the signal ratio plot (normalize MPAO, Full average and Full single by the system correction value), we can see that Full average is mostly ineffective as a result of the inhomogeneous refractive index distribution. The Full average signal is comparable to that of system correction. In certain regions, it is slightly lower than that of system correction. Full single is useful for the central segment and provides similar performance as MPAO, which is expected as the Full single wavefront is the same as the central segment wavefront used in MPAO. However, the performance of Full single can be worse than system correction for the surrounding segments. For these reasons, we consistently compare MPAO with system correction to demonstrate the benefit of MPAO.

## Supplementary Note 4, Adjusting the image segment size of MPAO for different biological systems

In the current design, we employed MPAO to simultaneously correct nine regions, covering a FOV of 450 × 450 $\mu m^2$ with 150 × 150 $\mu m^2$ within each image segment. For different biological systems, we would need to adjust the image segment size depending on the tissue heterogeneity. For example, we would need to use small image segments for correcting highly spatially varying aberration. There are a few straightforward solutions to make MPAO system flexible.

1. We can change the objective lens for different biological systems. For example, by changing from the Nikon 25× NA 1.1 (8 mm focal length) to Zeiss 40× NA 1.2 (4 mm focal length), we can effectively reduce the segment size by half to 75×75 $\mu m^2$ to accommodate more abrupt aberration changes.
2. Another strategy is to supply an abundant number of segments. For example, with 600 × 600 SLM pixels, we can sufficiently support 6×6 segments (100 × 100 SLM pixels per segment) to simultaneously correct 36 regions. While this could be unnecessary for less position dependent aberration, it can provide the versatility to supply finer aberration correction when needed.
3. We can also make prisms of different sizes and position them with a turret, which will allow the user to easily switch between different image segment (prism) sizes.